

4. Discuss mean and median with example.
5. What are uses of regression analysis.
6. Discuss the application of skewness.
7. What do you mean by conditional probability.

Group – C

(Long-answer Type Questions)

Answer any **three** questions of the following :

$$10 \times 3 = 30$$

8. Discuss the normal distribution and its properties.
9. Describe measure of dispersion and their types with examples.
10. Obtain mean and variance of Binomial distribution.
11. Explain moment generating function and its uses.
12. Discuss the application of Statistics in computer science.



HB – 14/2 (400)

(4)

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2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **all** the Groups as directed.*

Group – A

(Objective Type Questions)

(Compulsory)

1. Choose the correct answer of the following :

$$2 \times 10 = 20$$

- (a) Which distribution has bell shaped distribution ?

(i) Normal

(ii) Poisson

(iii) Binomial

(iv) None of these

HB – 14/2

(Turn over)

(b) The probability of an event lie between :

- (i) 0 to 1 (ii) - 1 to 0
- (iii) 0 to ∞ (iv) - 1 to 1

(c) In case of Poission distribution :

- (i) Mean = Variance
- (ii) Mean > Variance
- (iii) Mean < Variance
- (iv) Mean = 2 Variance

(d) r^2 is equal to :

- (i) $b_{xy} b_{yx}$ (ii) $b_{xy} + b_{yx}$
- (iii) $b_{xy} - b_{yx}$ (iv) b_{xy} / b_{yx}

(e) Median is equal to :

- (i) Q_2 (ii) Q_1
- (iii) Q_3 (iv) Q_4

(f) The mean of x_1 and x_2 is $(x_1 + x_2) / 2$. Then mean of $c x_1$ and $c x_2$ is :

- (i) $(c x_1 + x_2) / 2$ (ii) $c(x_1 + x_2) / 2$
- (iii) $(x_1 + c x_2) / 2$ (iv) $(x_1 + x_2) / 2$

(g) If A and B are not independent events then :

- (i) A' and B are not independent

(ii) A' and B' are independent

(iii) A and B' are not independent

(iv) None of these

(h) The correlation coefficient between price and demand is :

- (i) Positive (ii) Negative
- (iii) Linear (iv) Non-linear

(i) Variance is equal to :

- (i) $\sqrt{S. D.}$ (ii) $(S. D.)^2$
- (iii) S. D. (iv) 2S. D.

(j) The mean of first 'n' natural numbers (1, 2, 3 n) is :

- (i) $(n - 1) / 2$ (ii) $(n + 1) / 2$
- (iii) $n / 2$ (iv) n

Group – B

(Short-answer Type Questions)

Answer any **four** questions of the following :

5×4 = 20

2. Discuss the coefficient of variations.
3. Discuss Poission distribution.